

Practice Set 2

1. Write a program that takes a student's score as input and calculates their grade. Use the following grading system:

90-100: A

80-89: B

70-79: C

60-69: D

0-59: F

```
#without parameter
def calculate_grade():
    score = int(input("Enter student score: "))

    if score >= 90 and score <= 100:
        print("Grade: A")
    elif score >= 80 and score <= 89:
        print("Grade: B")
    elif score >= 70 and score <= 79:
        print("Grade: C")
    elif score >= 60 and score <= 69:
        print("Grade: D")
    elif score >= 0 and score <= 59:
        print("Grade: F")
    else:
        print("Invalid score")

#with parameter
def calculate_grade(score):
    if score >= 90 and score <= 100:
        print("Grade: A")
    elif score >= 80 and score <= 89:
```

```

        print("Grade: B")
    elif score >= 70 and score <= 79:
        print("Grade: C")
    elif score >= 60 and score <= 69:
        print("Grade: D")
    elif score >= 0 and score <= 59:
        print("Grade: F")
    else:
        print("Invalid score")

```

2. Write a Python program to calculate simple interest. Ask the user for the principal amount, the rate of interest, and the time period. Calculate and display the simple interest

```

#without parameter
def simple_interest():
    p = int(input("Enter Principal Amount: "))
    r = int(input("Enter Rate of Interest: "))
    t = int(input("Enter Time (in years): "))

    si = (p * r * t) / 100
    print("Simple Interest =", si)

#with parameter

def simple_interest(p, r, t):
    si = (p * r * t) / 100
    print("Simple Interest =", si)

```

3. Create a Python program to calculate the Body Mass Index (BMI) of a person. Ask the user for their weight in kilograms and height in meters. Calculate and display their BMI.

```

#without parameter
def calculate_bmi():

```

```

weight = int(input("Enter weight in kilograms: "))
height = float(input("Enter height in meters: "))

bmi = weight / (height * height)
print("Your BMI =", bmi)

#with parameter
def calculate_bmi(weight, height):
    bmi = weight / (height * height)
    print("Your BMI =", bmi)

```

4. Create a program that takes a person's age as input and classifies them into different age groups. The program should consider the following age categories:
 0-12: Child
 13-19: Teenager
 20-64: Adult
 65 and above: Senior Citizen

```

#without parameter
def age_category():
    age = int(input("Enter age: "))

    if age >= 0 and age <= 12:
        print("Category: Child")
    elif age >= 13 and age <= 19:
        print("Category: Teenager")
    elif age >= 20 and age <= 64:
        print("Category: Adult")
    elif age >= 65:
        print("Category: Senior Citizen")
    else:
        print("Invalid age")

#with parameter

```

```
def age_category(age):
    if age >= 0 and age <= 12:
        print("Category: Child")
    elif age >= 13 and age <= 19:
        print("Category: Teenager")
    elif age >= 20 and age <= 64:
        print("Category: Adult")
    elif age >= 65:
        print("Category: Senior Citizen")
    else:
        print("Invalid age")
```

5. Create a Python program to calculate the final price of an item with a discount. Ask the user for the original price and the discount percentage. Calculate and display the final price.

#without parameter

```
def final_price():
    price = float(input("Enter original price: "))
    discount = int(input("Enter discount percentage: "))

    discount_amount = (price * discount) / 100
    final = price - discount_amount

    print("Final Price =", final)
```

#with parameter

```
def final_price(price, discount):
    discount_amount = (price * discount) / 100
    final = price - discount_amount

    print("Final Price =", final)
```

6. Write a Python program that checks if a year is a leap year or not. Ask the user for a year and display whether it's a leap year or not.

```
#without parameter
def leap_year():
    year = int(input("Enter a year: "))

    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
        print(year, " It is a Leap Year")
    else:
        print(year, " It is not a Leap Year")

#with parameter
def leap_year(year):
    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
        print(year, " It is a Leap Year")
    else:
        print(year, " It is not a Leap Year")
```

7. Write a Python program that compares Three numbers and determines which one is greater or if they are equal. Ask the user for three numbers and display the result.

```
#without parameter
def compare_numbers():
    a = int(input("Enter first number: "))
    b = int(input("Enter second number: "))
    c = int(input("Enter third number: "))

    if a > b and a > c:
        print("Greatest number is:", a)
    elif b > a and b > c:
        print("Greatest number is:", b)
```

```

elif c > a and c > b:
    print("Greatest number is:", c)
else:
    print("All three numbers are equal")

#with parameter
def compare_numbers(a, b, c):
    if a > b and a > c:
        print("Greatest number is:", a)
    elif b > a and b > c:
        print("Greatest number is:", b)
    elif c > a and c > b:
        print("Greatest number is:", c)
    else:
        print("All three numbers are equal")

```

8. Write a Python program that checks if the given sides can form a valid triangle. Ask the user for the lengths of three sides and determine if they can create a triangle based on the triangle inequality theorem.

```

#without parameter
def check_triangle():
    a = float(input("Enter first side: "))
    b = float(input("Enter second side: "))
    c = float(input("Enter third side: "))

    if a + b > c and a + c > b and b + c > a:
        print("The sides form a valid triangle")
    else:
        print("The sides do not form a valid triangle")

#with parameter
def check_triangle(a, b, c):
    if a + b > c and a + c > b and b + c > a:
        print("The sides form a valid triangle")

```

```
else:  
    print("The sides do not form a valid triangle")
```